

🔍 Quick Comparison Table: Logical Functions in Excel

Function	Purpose	Works With	Best For	Limitations
 AND	Multiple conditions must be TRUE	Numbers, Text, Logical	Checking if all conditions are met	✗ Only returns TRUE/FALSE, no custom output
 OR	At least one condition must be TRUE	Numbers, Text, Logical	Checking if any condition is met	✗ Only returns TRUE/FALSE, no custom output
 XOR	Returns TRUE if only one condition is TRUE	Numbers, Text, Logical	Checking exclusive conditions	✗ If both conditions are TRUE/FALSE, returns FALSE
 IF	One-by-one condition check	Numbers, Text, Logical	Simple True/False checks	✗ Nested IFs get complex & slow
 SWITCH	Fixed values ke basis pe result dena	Text, Numbers	Replacing multiple IFs (like VLOOKUP alternative)	✗ Doesn't support greater/less than (>, <)
 LET	Variables define karne ke liye	Any formula	Reducing formula repetition & making sheets faster	✗ Only works inside Excel 365/2021, learning curve hai
 IFS	Multiple IF conditions ko simplify karna	Numbers, Text, Logical	Nested IFs ka better alternative	✗ Doesn't work well if default value required
 NOT	Condition ka opposite return karna	Logical values	Reversing TRUE/FALSE results	✗ Can't handle complex conditions alone

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 Formula	 Purpose	 Example	 Works With	 Limitations
 SUM	Numbers ka total nikalna	=SUM(A1:A5) → (A1 to A5 ka sum)	Numbers	✗ Text values ignore karta hai
 SUBTRACT (-)	Ek number ko dusre se minus karna	=A1 - A2	Numbers	✗ Directly SUBTRACT function nahi hota
 MULTIPLY (*)	Numbers ko multiply karna	=A1 * A2	Numbers	✗ Text values se error aata hai
 DIVIDE (/)	Numbers ko divide karna	=A1 / A2	Numbers	✗ #DIV/0! error aata hai agar denominator 0 ho
 AVERAGE	Numbers ka average nikalna	=AVERAGE(A1:A5)	Numbers	✗ Text values ignore hoti hain
 MIN	Sabse chhoti value dhoondna	=MIN(A1:A5)	Numbers	✗ Empty cells ignore karta hai
 MAX	Sabse badi value dhoondna	=MAX(A1:A5)	Numbers	✗ Empty cells ignore karta hai
 MOD	Remainder dhoondna	=MOD(10,3) → (10 ko 3 se divide karke remainder 1)	Numbers	✗ Negative values pe different behavior
 POWER	Number ka power nikalna	=POWER(2,3) → (2 ka cube = 8)	Numbers	✗ Negative powers me fractional output aata hai

 SQRT	Square root nikalna	=SQRT(25) → (5)	Numbers	✗ Negative numbers pe error
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Quick Comparison Table: Statistical Formulas in Excel

 Formula	 Purpose	 Example	 Works With	 Limitations
 AVERAGE	Numbers ka average (Mean) nikalna	=AVERAGE(A1:A5)	Numbers	✗ Text values ignore hoti hain
 MEDIAN	Middle value dhoondna	=MEDIAN(A1:A5)	Numbers	✗ Even number of values me midpoint nikalta hai
 MODE	Most frequently occurring value dhoondna	=MODE(A1:A10)	Numbers	✗ Multiple modes hone par sirf ek return karta hai
 COUNT	Kitni numeric values hain, count karta hai	=COUNT(A1:A10)	Numbers	✗ Text values count nahi hoti
 COUNTA	Numeric + Text dono values count karna	=COUNTA(A1:A10)	Numbers, Text	✗ Blank cells ignore nahi hota
 COUNTBLANK	Blank cells count karna	=COUNTBLANK(A1:A10)	Empty cells	✗ Non-empty cells ignore hoti hain
 MIN	Sabse chhoti value dhoondna	=MIN(A1:A10)	Numbers	✗ Text values ignore karta hai
 MAX	Sabse badi value dhoondna	=MAX(A1:A10)	Numbers	✗ Text values ignore karta hai

 Formula	 Purpose	 Example	 Works With	 Limitations
 STDEV.P	Population standard deviation nikalna	=STDEV.P(A1:A10)	Numbers	✗ Sample data ke liye STDEV.S use karein
 STDEV.S	Sample standard deviation nikalna	=STDEV.S(A1:A10)	Numbers	✗ Population data ke liye STDEV.P use karein
 VAR.P	Population variance nikalna	=VAR.P(A1:A10)	Numbers	✗ Sample data ke liye VAR.S use karein
 VAR.S	Sample variance nikalna	=VAR.S(A1:A10)	Numbers	✗ Population data ke liye VAR.P use karein
 RAND	0 aur 1 ke beech random number generate karna	=RAND()	Numbers	✗ Har bar sheet refresh hone par number change ho jata hai
 RANDBETWEEN	Fixed range ke beech random number dena	=RANDBETWEEN(1,100)	Numbers	✗ Har bar sheet refresh hone par value change ho jati hai

🔍 Quick Comparison Table: Lookup Formulas in Excel

📌 Formula	✅ Purpose	📊 Example	⚡ Works With	🚫 Limitations
 VLOOKUP	Vertical lookup (Ek column me value dhoondhna)	=VLOOKUP(101, A2:D10, 2, FALSE)	Table ke left-most column	❌ Sirf left se right search karta hai
 HLOOKUP	Horizontal lookup (Ek row me value dhoondhna)	=HLOOKUP(101, A1:Z2, 2, FALSE)	Table ke top-most row	❌ Sirf upar se neeche search karta hai
 XLOOKUP	VLOOKUP aur HLOOKUP ka advanced version	=XLOOKUP(101, A2:A10, B2:B10)	Left, right, upar, neeche sab possible	❌ Excel 2019 & earlier versions me nahi hai
 INDEX	Ek cell ki value row/column number se find karna	=INDEX(A2:C5, 2, 3)	Rows & Columns	❌ Direct lookup nahi karta
 MATCH	Kisi value ka position number find karna	=MATCH(50, A1:A10, 0)	Sorted & Unsorted data	❌ Exact match ke liye 0 use karna padta hai
 INDEX + MATCH	VLOOKUP ka advanced alternative	=INDEX(B2:B10, MATCH(101, A2:A10, 0))	Rows & Columns	❌ Thoda complex formula hai
 CHOOSE	Multiple options me se ek value return karna	=CHOOSE(2, "Apple", "Banana", "Mango")	Fixed data list	❌ Lookup table support nahi karta
 OFFSET	Dynamic cell range return karna	=OFFSET(A1, 2, 1)	Rows & Columns	❌ Complex aur volatile formula hai
 FILTER	Multiple matching values return karna	=FILTER(A2:A10, B2:B10="India")	Dynamic Arrays	❌ Excel 2019 & earlier me nahi hai

Text Formulas in Excel

Formula	Usage	Example	Result
LEFT 	Extracts first 'n' characters from text	=LEFT("Excel", 2)	Ex
RIGHT 	Extracts last 'n' characters from text	=RIGHT("Excel", 2)	el
MID 	Extracts characters from the middle	=MID("Excel", 2, 3)	xce
LEN 	Returns length of text	=LEN("Excel")	5
TRIM 	Removes extra spaces	=TRIM(" Hello World ")	Hello World
LOWER 	Converts text to lowercase	=LOWER("Excel")	excel
UPPER 	Converts text to uppercase	=UPPER("Excel")	EXCEL
PROPER 	Capitalizes first letter of each word	=PROPER("hello world")	Hello World
CONCAT 	Combines text (New Formula)	=CONCAT(A1, B1)	HelloWorld
TEXTJOIN 	Combines text with a separator	=TEXTJOIN(" ", TRUE, A1, B1, C1)	Hello World Excel
FIND 	Finds position of a character (Case-Sensitive)	=FIND("c", "Excel")	3
SEARCH 	Finds position of a character (Case-Insensitive)	=SEARCH("C", "Excel")	3
SUBSTITUTE 	Replaces text in a string	=SUBSTITUTE("Hello World", "World", "Excel")	Hello Excel

Formula	Usage	Example	Result
REPT 	Repeats text multiple times	=REPT("A", 5)	AAAAA

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🕒 Date & Time Formulas in Excel

Formula	Usage	Example	Result
TODAY 📅	Returns the current date	=TODAY()	2025-02-25
NOW ⌚	Returns the current date & time	=NOW()	2025-02-25 14:30
DATE 📅	Creates a date from year, month, day	=DATE(2025, 2, 25)	2025-02-25
TIME ⌚	Creates a time from hours, minutes, seconds	=TIME(14, 30, 0)	14:30:00
YEAR 📅	Extracts year from a date	=YEAR(A1) (A1 = 25-Feb-2025)	2025
MONTH 📅	Extracts month from a date	=MONTH(A1) (A1 = 25-Feb-2025)	2
DAY 📅	Extracts day from a date	=DAY(A1) (A1 = 25-Feb-2025)	25
HOUR ⌚	Extracts hour from time	=HOUR(A1) (A1 = 14:30:00)	14
MINUTE ⌚	Extracts minutes from time	=MINUTE(A1) (A1 = 14:30:00)	30
SECOND ⌚	Extracts seconds from time	=SECOND(A1) (A1 = 14:30:45)	45
EDATE ➕ 📅	Adds months to a date	=EDATE(A1, 3) (A1 = 25-Feb-2025)	25-May-2025
EOMONTH ⬅️ 📅 END	Finds last day of a month	=EOMONTH(A1, 0) (A1 = 25-Feb-2025)	28-Feb-2025
DATEDIF 📅	Calculates difference between dates	=DATEDIF(A1, B1, "Y") (Years between two dates)	2
NETWORKDAYS 🏢	Counts working days between two dates	=NETWORKDAYS(A1, B1)	20

Formula	What It Does	Example
PV (Present Value) 	Aaj ke time par kisi future investment ki value kya hai	=PV(5%/12, 60, -5000, 0, 0) → ₹2,59,664
FV (Future Value) 	Future me ek investment ki expected value kya hogi	=FV(5%/12, 60, -5000, 0, 0) → ₹3,48,850
PMT (Loan Payment) 	Loan ka har mahine ka installment kitna hoga	=PMT(5%/12, 60, -200000, 0, 0) → ₹3,774
RATE (Interest Rate) 	Loan ya investment ka interest rate nikalna	=RATE(60, -3774.25, 200000, 0, 0) → 5%
NPV (Net Present Value) 	Ek investment ka total profit ya loss check karna	=NPV(10%, A1:A5) → ₹5,000
IRR (Internal Rate of Return) 	Ek project ya investment ka expected return	=IRR(A1:A5) → 12.5%
CUMIPMT (Total Interest Paid) 	Pure loan tenure me total kitna interest diya jayega	=CUMIPMT(5%/12, 60, 200000, 1, 60, 0) → ₹26,654
CUMPRINC (Total Principal Paid) 	Pure loan tenure me total principal amount kitna diya gaya	=CUMPRINC(5%/12, 60, 200000, 1, 60, 0) → ₹1,73,345
SLN (Straight-Line Depreciation) 	Asset har saal kitna value lose karega (fix depreciation)	=SLN(10000, 1000, 5) → ₹1,800 per year
DB (Declining Balance Depreciation) 	Pehle zyada aur baad me kam depreciation (accelerated)	=DB(10000, 1000, 5, 1) → ₹2,000 first year

Information Formulas in Excel (Simplified)

Formula	What It Does	Example
ISNUMBER 	Check karta hai ki cell me number hai ya nahi	=ISNUMBER(A1) → TRUE / FALSE
ISTEXT 	Check karta hai ki cell me text hai ya nahi	=ISTEXT(A1) → TRUE / FALSE
ISBLANK 	Check karta hai ki cell khali hai ya nahi	=ISBLANK(A1) → TRUE / FALSE
ISERROR 	Check karta hai ki cell me koi error hai ya nahi	=ISERROR(A1) → TRUE / FALSE
ISNA 	Check karta hai ki cell me #N/A error hai ya nahi	=ISNA(A1) → TRUE / FALSE
ISLOGICAL  	Check karta hai ki cell me TRUE/FALSE value hai ya nahi	=ISLOGICAL(A1) → TRUE / FALSE
ISFORMULA 	Check karta hai ki cell me formula hai ya nahi	=ISFORMULA(A1) → TRUE / FALSE
TYPE 	Data ka type (1=Number, 2=Text, 4=Logical) return karta hai	=TYPE(A1) → 1 / 2 / 4
N 	Logical value ya text ko number me convert karta hai	=N(TRUE) → 1
ERROR.TYPE 	Agar cell me error hai toh uska error code return karta hai	=ERROR.TYPE(A1) → 1 to 8 (error codes)
INFO 	System-related information deta hai	=INFO("osversion") → "Windows 10"

Simple Database Formulas in Excel

Formula	Use Kaha Hota Hai?	Example
DAVERAGE 	Kisi table me ek condition ke basis par average nikalta hai	=DAVERAGE(A1:D10, "Salary", F1:F2) → Salary ka average jo F1:F2 condition match karega
DCOUNT 	Table me numbers count karta hai jo condition match karein	=DCOUNT(A1:D10, "Age", F1:F2) → Jo rows F1:F2 condition match karti hain, unke age count karega
DCOUNTA 	Non-empty cells (text + numbers) count karta hai	=DCOUNTA(A1:D10, "Name", F1:F2) → Names count karega jo condition se match karte hain
DGET 	Table me se ek specific value nikalta hai	=DGET(A1:D10, "Salary", F1:F2) → Jo row condition match karegi, uska salary value return karega
DMAX 	Table me sabse bada (max) value nikalta hai	=DMAX(A1:D10, "Salary", F1:F2) → Salary ka max value return karega jo condition match karega
DMIN 	Table me sabse chhota (min) value nikalta hai	=DMIN(A1:D10, "Salary", F1:F2) → Salary ka min value return karega jo condition match karega
DPRODUCT 	Condition match karne wale numbers ka product karta hai	=DPRODUCT(A1:D10, "Sales", F1:F2) → Sales values ka multiplication karega jo condition match karegi
DSTDEV 	Sample standard deviation nikalta hai (variation check karta hai)	=DSTDEV(A1:D10, "Sales", F1:F2) → Sales ki variation (spread) batata hai
DSTDEVP 	Pure population ka standard deviation batata hai	=DSTDEVP(A1:D10, "Sales", F1:F2) → Poore sales data ka spread dekhta hai
DSUM 	Table me condition ke basis pe sum karta hai	=DSUM(A1:D10, "Sales", F1:F2) → Sales ka total jo condition match karega

Formula	Use Kaha Hota Hai?	Example
DVAR 	Sample variance (data ka spread) batata hai	=DVAR(A1:D10, "Sales", F1:F2) → Sirf sample data ka spread check karega
DVARP 	Poore data ka variance batata hai	=DVARP(A1:D10, "Sales", F1:F2) → Pure sales data ka spread check karega

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